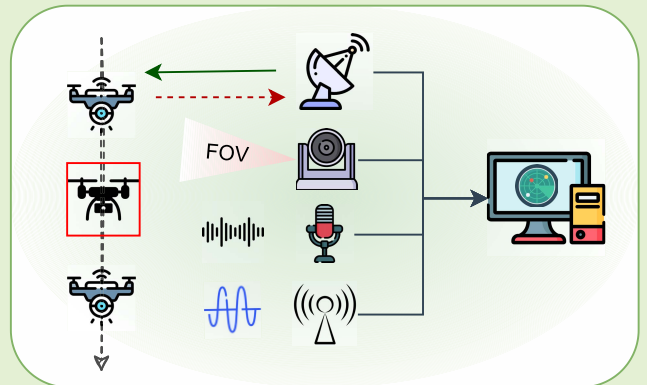


On the Detection of Unauthorized Drones—Techniques and Future Perspectives: A Review

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Abstract—The market size of civilian drones is tremendously increasing and is expected to reach 1.66 million by the end of 2023. The increase in number of civilian drones poses several privacy and security threats. To safeguard critical assets and infrastructure and to protect privacy of people from the illegitimate uses of commercial drones, a drone detection system is inevitable. In particular, there is a need for a drone detection system that is efficient, accurate, robust, cost-effective and scalable. Recognizing the importance of the problem, several drone detection approaches have been proposed over time. However, none of these provides sufficient performance due to the inherited limitations of the underlying detection technology. More specifically, there are trade-offs among various performance metrics e.g., accuracy, detection range, and robustness against environmental conditions etc. This motivates an in-depth study and critical analysis of the existing approaches, highlighting their potential benefits and limitations. In this paper, we provide a rigorous overview of the existing drone detection techniques and a critical review of the state-of-the-art. Based on the review, we provide key insights on the future drone detection systems. We believe these insights will provide researchers and practicing engineers a holistic view to understand the broader context of the drone detection problem.

Index Terms—Unmanned aerial vehicles, drone detection, privacy, radio, radar, security, visual.



I. INTRODUCTION

THE Unmanned Aerial Vehicles (UAVs) commonly known as drones are becoming an integral part of our lives due to their widespread deployment in various applications [1]. Initially linked with military applications, drones enjoy an ever-increasing popularity in civil applications. The drones' application landscape is quite diverse and spans across sev-

eral verticals including entertainment (aerial photography and videography), geology (mapping and surveying), transportation (traffic monitoring), security (search and rescue, crowd monitoring, emergency response, and disaster relief), shipping (parcel delivery), agriculture (crop monitoring, spraying, and bird control), and communication (emergency communication infrastructure). These unprecedented applications and use cases of civilian drones can revolutionize the society by enabling a more autonomous world. Considering the aforementioned applications, one can expect the future with swarms of drones flying in the air like flocks of birds. While the current regulations on the use of drones in civil applications restrict their growth to some extent, their adoption will still be increasing over time. The Federal Aviation Administration (FAA), USA forecasts the number of model drones¹ reaching a total number of 1.39-1.66 million and non-model (commercial) drones reaching a total number of 835,000 in the United States (US) [3].

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¹According to FAA, model aircraft or drones “do not interfere with manned aircraft, be flown within sight of the operator (below 400 feet in uncontrolled airspace.) and be operated only for hobby or recreational purposes” [2].

The anticipated increase in the number of civilian drones flying in the airspace raises huge concerns and is often considered as a threat to public privacy and security. Being “unmanned”, drones can be anonymously used for many illegitimate activities such as spying, hacking, target attacks etc. We describe few of these threats in specific scenarios (as depicted in Fig. 1) to highlight the motivation behind the need of a drone detection system.

A. Potential Threats of Commercial Drone's

1) *Breaching No-Fly Zone (NFZ)*: No-Fly Zone (NFZ) is a restricted area where its owner does not want any drone to fly into [4]. The technique to restrict unauthorized drones to enter an NFZ is referred to as *geofencing*. An NFZ can be fixed and predefined or it can be variable and dynamically identified [5]. Typical examples of NFZs are airports, metro stations, oil and gas facilities, state's residencies and offices of intelligence agencies. Although some UAV manufacturers implement NFZ definitions in their firmware (e.g., DJI and Parrot [6]) to restrict drones' entry into these areas, these can be easily altered by hackers to use them for illegitimate access to NFZs.

2) *Privacy Invasion*: The proliferation of consumer drones can be threatening to individual privacy [7], [8]. With a number of consumer drones for legal activities such as filming, food and parcel delivery, it is inevitable that a drone equipped with a camera can be spying on someone sitting in the backyard of his (or her) house. It becomes challenging to distinguish between drones (authorized to fly in the region) for legitimate and illegitimate purposes as the spying activity depends on the orientation of the drone's camera rather than the location [9].

3) *Cyber-Attacks*: Drone-based cyber-attacks [10], [11] yet provide another strong motivation for drone detection and geofencing. Traditionally, enterprises have been protecting their networks and computer systems by deploying local private networks. However, as the use of wireless broadband (e.g., Wi-Fi) and cloud-based storage and computing is becoming ubiquitous, physical proximity restrictions become more crucial to prevent cyber-attacks [12]. In order to get access to the Wi-Fi or other short range communication network, hackers need to move into the range of these networks and drones provide them this opportunity.

4) *Terrorism and Target Attacks*: In the past, drones have been used to carry out terrorist attacks on public figures in several parts of the world. Drones not only enable terrorists to enter restricted areas to target a victim but also provide them the opportunity to remotely track the target on the way when they are driving. As such, unfortunately drones have become attractive tools for target attacks [13], [14]. Several events in the near past highlight this challenge. The recent drone attack on Saudi airport in Jazan [15], and another attack on Russian's airbase and naval bases by a swarm of miniature drones armed with explosives [16], are only a few examples of the use of drones in terrorism.

The aforementioned examples emphasize the significance of drone detection to prevent their illegitimate uses. Hence, this topic received great attention from both industry and academia in the last few years. However, unauthorized drone

detection is a challenging problem. As an early approach to stop the misuse of consumer drones, some drone manufacturers introduced special override control commands to restrict the drone's entry into one or more NFZs. However, this rarely helps when the drone is intentionally being used for unlawful activities as such firmware commands can be easily altered by hackers. Secondly, although defining and implementing NFZ capabilities by drone manufacturers can help to restrict unintentional drone entry into restricted areas, not all manufacturers implement NFZ definitions into their products. Hence, having a stand-alone drone detection system becomes the obvious solution in several scenarios especially where drones can pose as a threat to security and privacy of people, assets, or infrastructure.

A drone detection system can use one or more sensors to detect the presence of a drone. Existing methods for drone detection include sound-based (acoustic detection), Radio Frequency-based (RF detection), radar-based, and camera-based (visual detection). The objective of the system is to detect the presence of a drone and determine its location along with other information such as type of drone, its direction, speed, and capabilities.

B. Contribution

Although drone detection is a significant problem, the existing techniques for drone detection have several limitations in terms of accuracy, range, robustness, cost, and reliability. Indeed, it is an active area of research with significant amount of research work been contributed in the past few years. To summarize the significant amount of research and development in the area, there also exist a few magazine and survey articles. However, we found these studies either are not up to date or they miss several important aspects. For instance, some earlier works like [17] provides a review of radar-based drone detection, classification and radar cross section (RCS) characterisation methods. The works in [18]–[20] briefly explain the four basic drone detection approaches without discussing the state-of-the-art. Ref. [21] presents a review which is more focused on machine learning (ML)-based methods for drone detection with limited insights on the performance of original detection approaches and their real-world adoption. It also does not cover novel methods based on sensor fusion which is an interesting direction and a potentially more promising approach for drone detection. Similarly, [22] reviews drone detection, identification, and neutralization, with less emphasis on the detection techniques. The study does not cover the state-of-the-art in detection approaches and is more focused on the system design from regulatory perspective. Similarly, authors in [23] presented a detailed review of drone detection approaches which is more focused on the the applications of deep learning on multi-sensor data.

As compared to the existing surveys, in this survey we provide a holistic view of the drone detection problem, explicitly explaining the goals and various attributes of a drone detection system. First, the widely used approaches for drone detection are investigated highlighting the advantages, technical limitations and potentials for improvement. Then, a detailed critical review of the state-of-the-art covering recent developments

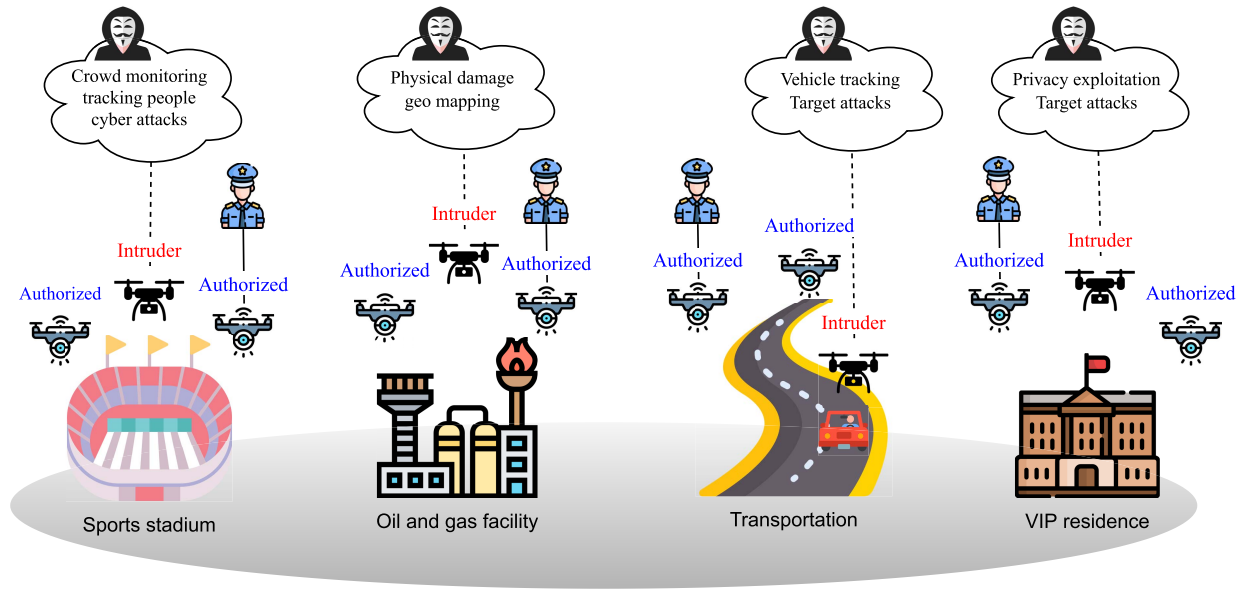


Fig. 1. Various drone scenarios showing drones deployed for legitimate uses with few intruder drones exist performing illegitimate activities (e.g., target tracking, cyber attacks, physical attacks, privacy attacks etc).

which are not covered in the previous studies. As the reader gets an overall understanding of these approaches, we present a review of novel techniques based on sensor fusion. Lastly, we provide the conclusion and clear directions and insights on future drone detection systems. We believe that such insights will be very useful for the novice researchers and practitioners to build more efficient solutions.

The paper is organized as follow:

Section II describes the architecture of Unmanned Aerial System (UAS) and illustrates the detection problem. Section III provides an indepth overview of the existing drone detection approaches. It also includes a critical review of these methods and highlight their potential advantages and inherited limitations. Lastly, section V presents conclusion and future insights. A list of acronyms sued in this paper is presented in Table I.

II. DRONE DETECTION ILLUSTRATED

In this section, we briefly illustrate the architecture of an Unmanned Aerial System (UAS) as the drone detection approaches rely on the fundamental UAS operations. Then, the drone detection problem and the key attributes of such systems are described to understand the broader context of the problem.

A. Architecture of UAS

The architecture of a typical UAV system (as shown in Fig. 2) comprises of two parts: (i) a UAV or drone, and (ii) a controller (also called remote controller or ground controller). A **drone** is typically equipped with a power source, an onboard controller, multiple sensors (including inertial sensors, magnetometer, gyroscope, Global Navigation Satellite System (GNSS) e.g., Global Positioning System (GPS) module, pressure sensor and visual sensors (i.e., camera)), and a wireless transceiver [24]. These sensors enable the drone

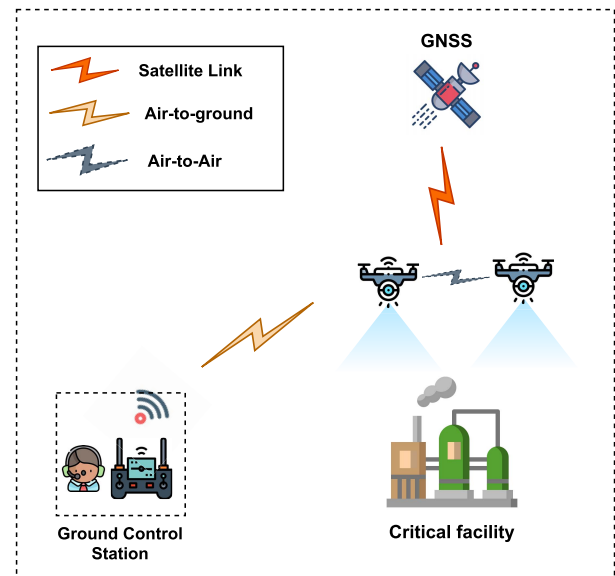


Fig. 2. Typical architecture of an unmanned aerial system (UAS).

to maintain smooth flight, safe landing, path following, and object detection. The **remote controller** can be manually operated or semi/fully automated control unit comprising of the control unit and wireless transceiver. The communication link between the UAV and the remote controller can use any wireless technology e.g., Wi-Fi, cellular (3G, 4G, or 5G) or Bluetooth. The RF signal transmitted over the communication link can be used to detect drones.

B. Drone Detection

Drone detection generally refers to detecting an intruder drone performing illegitimate activity (e.g., entering an NFZ or video recording etc.). Broadly speaking, detection involves determining the presence, identity, size, position, direction, speed, and capabilities of the intruder drone. In this work,

TABLE I
LIST OF ACRONYMS

Acronym	Meaning
AIB	Axially Integrated Bispectra
CA-CFAR	Cell Averaging Constant False Alarm Rate
CNN	Convolution Neural Network
CSI	Channel State Information
DA	Discriminant Analysis
DCF	Discriminative Correlation Filter
FD	Fractal Dimension
FMCW	Frequency Modulated Continuous Wave
GMM	Gaussian Mixture Models
GNSS	Global navigation Satellite System
GSVD	Generalized Singular Value Decomposition
HOG	Histogram of Oriented Gradients
IMU	Inertial Measurement Unit
IR	Infra Red
IVA	Independent Vector Analysis
KCF	Kernelized Correlation Filter
KNN	K Nearest Neighbors
LBF	Local Binary Patterns
LSTM	Long Short-Term Memory
LPCC	Linear Predictive Cepstral Coefficients
LOS	Line Of Sight
MFCC	Mel Frequency Cepstral Coefficient
NB	Naive Bayes
NFZ	No Fly Zone
PSD	Power Spectral Density
PSE	Power Spectral Entropy
PW3PD	Pixel-Wise Posterior 3D
RCS	Radar Cross-Section
RF	Random Forest / Radio Frequency
RMS	Root Mean Square
RNN	Recurrent Neural Network
SFS	Signal Frequency Spectrum
SIB	Square Integrated Bispectra
SRP-PHAT	Steered response Power with Phase Transform
SVM	Support Vector Machine
TDOA	Time Difference Of Arrival
TVD	Time Velocity Diagrams
UAS	Unmanned Aerial System
UAV	Unmanned Aerial Vehicle
WEE	Wavelet Energy Entropy

we refer “detection” as simply to detect the presence of a drone unless otherwise specified. Detecting the presence of a drone is a binary problem, whereas identification can be modeled as binary classification (e.g., whether a drone is authorised or unauthorised?) or multi-class classification (identification of a drone based on the make, model, activity or a unique ID).

Apart from detecting the presence and identification of a drone, there are other significant attributes that are often desired in a drone detection system. A list of such attributes is given in Table II.

III. DRONE DETECTION METHODS

Some countries have implemented certain regulatory measures to restrict the illegitimate uses of drones. For instance, some UAVs manufacturers (e.g., Da-Jiang Innovations (DJI)) adopted the approach to program drones such that to override any control command that directs it to enter an NFZ. However,

TABLE II
DRONE DETECTION PARAMETERS

Parameter	Description
Presence	A binary value to show presence or absence of a drone in the region of interest.
Type	A discrete ID of the drone's type (e.g., rotary wing, fixed wing, flapping wing)
Model	A unique ID or mapping of the drone's make and/or model.
Size	A discrete (e.g., size-1, size-2) or continuous (e.g., 0.1m, 0.5m etc.) value to give an estimated size of the drone.
Distance	A continuous value to measure the distance to the detected drone.
Height	A continuous value to measure vertical distance from a fixed reference point such as earth surface)
Speed	A continuous value to record drone's instantaneous or average speed.
Direction	Discrete value (e.g., north, north-east) or continuous value (angle) of the approaching drone.
Count	Number of drones (e.g., single, swarm)

these measures can only restrict drones' entry and do not solve the problem of preventing unauthorized drones operations in non-restricted areas causing several threats as discussed in the previous section. Nevertheless, these restrictions also do not fully protect NFZs from unauthorized drones' entry. First, adversaries can disable the override commands in the drone's firmware. Second, over the course of time, existing NFZs may change and new NFZs may be defined by regulators. Such changes are implemented as software updates to the drones being sold and used by consumers. However, users can typically disable the automatic downloading of new updates. Third, such regulatory compliance to implement NFZ-related overrides in the firmware require global consensus among drone manufacturers. Nevertheless, if such consensus is built in the industry, somebody can build their own drones without NFZs implementation.

The aforementioned discussion on the regulatory measures to prevent the illegitimate operations of commercial drones pave the way to develop a reliable standalone drone detection system. Such a detection system uses one or more drone's features to detect their presence, location or other attributes of interest. Typically, drones emit certain signals such as RF signals, sound waves, heat signals, and Doppler effects on the surrounding environment. Therefore, these signals can be captures using sensor systems and processed to detect the presence of a drone in a region of interest. Based on the adopted sensing method, drone detection systems are generally classified into four categories i.e., acoustic detection, RF detection, radar detection, and visual detection. However, more recently hybrid methods using sensor fusion and alternate novel methods have been proposed. Fig. 3 illustrates the drone detection methods covered in this paper.

Fig. 4 illustrates the architecture of the four basic approaches, namely acoustic detection, RF detection, radar detection, and visual detection.

A. Acoustic Detection

The rotating blades of drones produce distinct acoustic signals (sound waves) which can be used to detect and

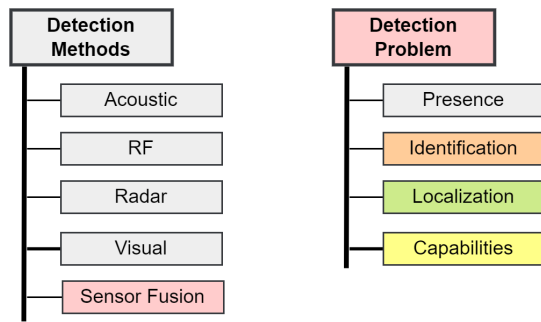


Fig. 3. Drone detection methods and detection attributes/problems.

identify a drone model. The sound is usually generated by different drone's components including propeller blades, engine, or wind [25]. However, the propeller blade sound has a relatively higher amplitude and is thus often used for detection. The acoustic signal is captured using highly sensitive acoustic sensors (microphones) and is processed to map with the drone's ID in a database. This process is known as *fingerprinting* [26]. Fig. 5 depicts few samples of acoustic signatures of two commercial drones extracted from a publicly available dataset [27].

The database typically contains the acoustic signatures (aka fingerprints) of all drones indexed by unique IDs of drones. The acoustic signals are processed either using traditional techniques or novel ML algorithms. Examples of traditional techniques include Multiple Signal Classification based on Generalized Singular Value Decomposition (GSVD-MUSIC) and Steered Response Power with Phase Transform (SRP-PHAT) [28]. These techniques are used to extract features such as linear predictive coding, Fourier coefficients, and Mel Frequency Cepstral Coefficients (MFCC). These features are compared against the pre-saved signatures to search for a match (i.e., ID of the UAV). Alternatively, ML algorithms are used to train a model using the database to automatically classify a new detected acoustic signal. Some commercial off-the-shelf systems using acoustic detection method include Discovair G2 [29], Drone Hound [30].

1) Advantages and Limitations: Detecting drones using acoustic signals can be relatively cheaper (although cost increases with the high accuracy requirement). It has the advantage to work unaffectedly in low light and diverse weather conditions such as fog or dust. It also does not need Line Of Sight (LOS) to the target drone. Acoustic systems are best suited to remote areas with no background noise. The accuracy of this method highly deteriorates in the presence of background noise and cease to detect if the drone uses sound-suppression system. The detection range is limited (ideally up to ~ 200 meter) and can vary according to the environment. Acoustic detection is not affected by the drone's speed, provided that the matching/prediction algorithm is efficient to detect drones in real-time. Furthermore, there is no study available whether the drone sound changes with long use or while its battery is low.

2) State-of-the-Art: The use of acoustic sensors in drones for object localization has been investigated in [31] and [32]. Acoustic detection schemes employ one or more microphones

to detect the sound waves. For instance, [33] uses a single microphone, [34] uses four microphones, whereas [35] uses 120 microphones. A single microphone typically offers poor detection; hence, more efficient detection systems use arrays of microphones. A simple acoustic drone detection system is presented in [36]. Audio signals from drones' engines are recorded in a studio environment. The recorded audio samples ($\sim 120 - 180$ seconds) are then processed to compute their normalized Power Spectral Densities (PSD). Then, using cross correlation of the normalized PSDs, the drones are identified based on their acoustic signatures (i.e., PSD). Using simple techniques such as cross correlation over PSD may not be very accurate; hence, authors in [37] propose the use of ML and more useful features including MFCC and Linear Predictive Cepstral Coefficients (LPCC). An extension of the work uses PSD, Root Mean Square (RMS) of the PSD, and MFCC extracted using Independent Component Analysis (ICA) and ML algorithms i.e., Support Vector Machine (SVM) [38] and K-Nearest Neighbors (KNN) [39] to detect drones [40]. A closely similar work propose the use of Independent Vector Analysis (IVA) for feature extraction from acoustic signals [41].

While standard ML techniques such as SVM and KNN for classification can provide reasonable performance, more recently DL methods [27], [33], [42]–[47] have been adopted to develop more accurate and robust detection model. Authors in [45] developed detection systems with a more advanced ML model using Long Short-Term Memory (LSTM) [48]. Moreover, authors collected audios of drones from online resources (i.e. YouTube) to enrich the training data for training the classifier model. To cope with the Doppler effect, authors applied two data augmentation techniques (amplitude scaling and frequency warping). In [27], authors compared the performance of Convolution Neural Network (CNN) [49] versus Recurrent Neural Network (RNN) [50] techniques for drone detection and classification using a dataset of acoustic signatures of two commercially available drone models by Parrot. Their results showed that CNN outperforms RNN for both drone models. Acoustic technique can be used for drone localization as well. In [33], authors uses SVM along with Time Difference Of Arrival (TDOA) to detect and localize drones. A comparative study on the effectiveness of several ML models for acoustic detection of drones is conducted in [42]. The authors compared three ML models i.e., Gaussian Mixture Models (GMM) [51], CNN, and RNN and showed that RNN outperform other models.

3) Summary and Insights: Acoustic detection of drones generally employs fingerprinting technique in which recorded sounds of drones are compared with pre-saved acoustic signatures. Using the raw samples from acoustic sensors is not an efficient approach; hence, useful features are extracted rather than using raw sound data. Some useful features being used in the literature are PSD, RMS of PSD, MFCC, and LPCC. Similarly, using simple matching techniques such as cross correlation may not provide good performance. ML-based classification techniques such as SVM, RF, and KNN can provide better accuracy. More recent studies adopted more sophisticated ML techniques. Traditional ML models can not

detect the temporal features and hence rely on extracted features. Deep Learning (DL) models such as LSTM can thus be utilized to detect the temporal variations in the acoustic signals. Another good technique is to use CNN for automatic feature extraction and detection.

Despite the performance gains by ML and DL approaches, acoustic detection is still at infancy. All the benchmarks presented in these works are based on limited experiments or and tested on only a few available datasets. Furthermore, a large portion of works in this category is focused on only detection (presence) of drones, few works on classification. There are rare works that provide localization of drones using acoustic detection. Furthermore, we did not found any study on the use of acoustic detection to detect simultaneously multiple drones.

B. RF-Based Detection

Drones use RF signals typically in the range of 2.4 GHz to 5 GHz [52], to communicate with the controller. Drones equipped with Wi-Fi uses 5.4 GHz while 5G drones operate at 3.5 GHz range. Other less common bands can be in the 1.2 GHz and 1.3 GHz range [53].² Thus an RF sensor (or scanner) can be used to passively listen to the signals transmitted between a drone and its controller. There are two methods of RF signal scanning. In the first method, the receiver listens and decodes the packets being transmitted between a drone and its controller. As the full packets are decoded, this method provides more detailed information such as drone's identity, location, speed and even data inside the packets such as video streams. However, this method requires network sniffing, which is not legal in many countries. In the second method, the RF receiver does not decode the packets but only captures the RF signals (consists of amplitude and phase) and finds specific patterns in the captured data. This method provides limited information i.e., it detects drone's presence, predicts drone's make and model, and estimate the drone's location (e.g., the direction and distance from the receiver). Due to the regulatory constraint, the second method is mostly used for drone's detection.

As the frequency used by the drone is usually not known, an RF scanner hops between several frequencies and listens to the communication in order to cover all the frequency ranges that a drone and its controller might operate at [54]. The received radio signal by the RF scanner is analyzed to detect features e.g., the vibration effects on the RF signal strength due to drone's propellers. Another approach is to scan specific bands used by the drones. For instance, most controllers and drones exchange signals 30 times every second. Therefore, the signal laying under the 100Hz band can be analyzed to search for certain distinctive features. Fig. 6 depicts sample RF signatures of a commercial drone, extracted from a publicly available dataset [55]. Some commercial off-the-shelf systems using RF detection method include the popular DJI Aeroscope [56], DroneDefense [56], AirWarden [57].

1) Advantages and Limitations: RF-based detection is currently the most widely deployed method for detecting drones. The method is typically more robust to weather conditions and does not require LOS between the RF receiver and the target drone. RF-based detection can also localize drones and its operators. Despite the wide use and benefits, RF-based detection has some limitations. On the top, the method can not detect autonomous drones (drones which do not communicate with any controller). Their performance is highly impacted in the presence of interference from other RF sources. The performance also deteriorates in the presence of rugged terrain with obstacles. RF detection can typically provide drone's direction in the range of 45° and a general sense of distance rather than accurate information. Hence, the impact of drone's speed on RF detection is minimum.

2) State-of-the-Art: Earlier works on RF detection of drones use statistical features extracted from the captured RF signal to detect and classify drones. For instance, authors in [58] and [59] propose to employ a wireless packet sniffer to capture the transmission of drones and extract statistical features to identify drones. Similarly, [60] propose a passive drone detection and classification system that uses ML algorithms i.e., KNN, Discriminant Analysis (DA), SVM, and Neural Networks (NN). The features set is reduced using Neighborhood component analysis (NCA) to improve the model accuracy and robustness. A Wi-Fi based drone detection and identification system leveraging ML is proposed in [61]. The RF sensor monitor the channel to capture the Channel State Information (CSI) information. The CSI data is processed to extract features such as Signal Frequency Spectrum (SFS), the Wavelet Energy Entropy (WEE), and the Power Spectral Entropy (PSE). ML techniques including SVM, Random Forest (RF) [62], Naive Bayes (NB), Ensemble Learning (EL), and KNN are then used to detect the drone. The authors in [61] also proposed in [63] to extract other features from CSI including Fractal Dimension (FD), Axially Integrated Bispectra (AIB), and Square Integrated Bispectra (SIB) and then applied PCA [64] and NCA [65] methods to reduce the feature dimensionality and trained ML classifiers (SVM, KNN) to perform the detection task. The aforementioned works used traditional ML algorithms, which typically does not provide sufficient accuracy and robust performance in practical large deployments due to the non-linearity in the RF signals. Hence, more sophisticated Deep Learning methods are recently adopted. Authors in [66] conducted a feasibility study on the use of DL to detect and identify drones. A dataset (i.e., DroneRF) of various drones' sounds is recorded and processed to fed to the DL models. A similar work using the dataset developed in [66] is found in [67] and [68] for drone identification using DNN and residual-CNN respectively. CNNs are well-known for their performance due to automatic feature extraction capabilities; however, they do not capture the temporal dependencies in the signal. Typically used DL techniques for time-varying data such as RF signals include RNN and LSTM. An Deep RNN (DRNN) model trained over the DroneRF dataset [66] is proposed in [69]. The trained model is deployed over multiple edge servers for drone identification. Authors in [70] propose a

²IEEE frequency bands: L (1.0 to 2.0 GHz), S (2.0 to 4.0 GHz), C (4.0 to 8.0 GHz).

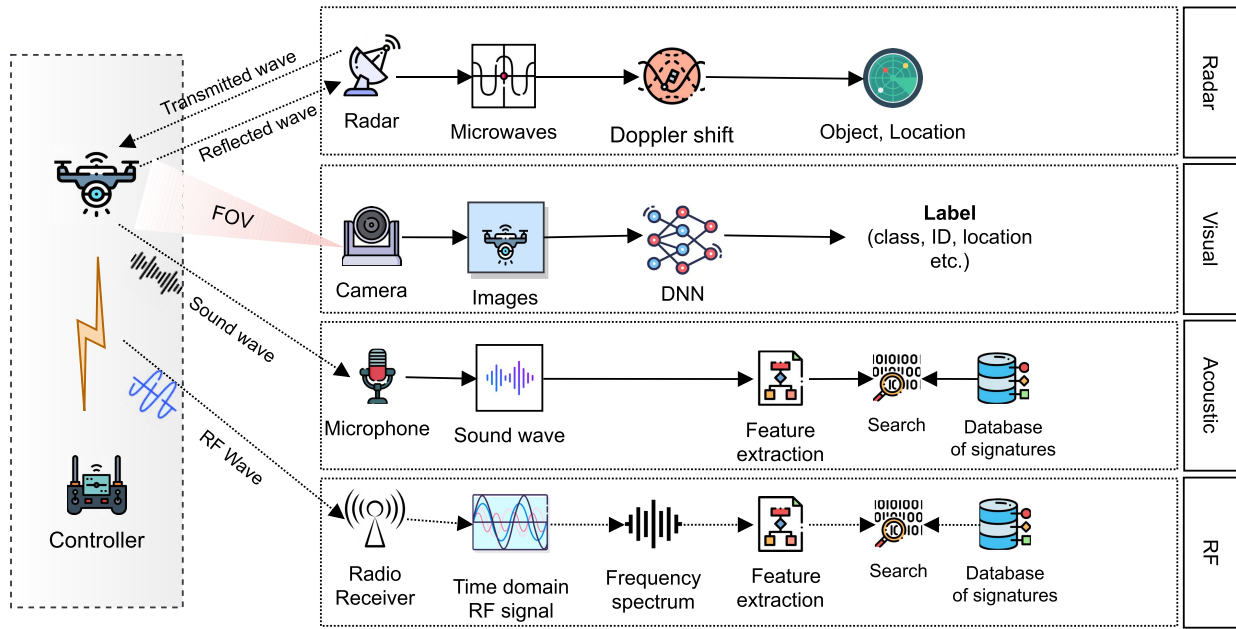


Fig. 4. Typical drone detection approaches (i.e., RF, acoustic, radar, and visual detection schemes).

multi-channel DNN (MC-DNN) architecture for drone detection and classification by reducing the classification errors caused by the similarities among categories of RF UAV signals. Three methods (i.e., (i) data truncation and normalization, (ii) moving average filter, and (iii) concatenation) are used to improve the classification accuracy.

3) Summary and Insights: RF detection provides various advantages over other methods such as no LOS requirement, longer detection range, robustness to weather conditions and better accuracy. The primary concern in its adoption is that it does not work on autonomous drones. Autonomous drones are likely to be used in the future UAS systems, however the major portion of future drones is expected to be supervised by ground controllers. Thus, this limitation would not be applicable to most scenarios. To measure the true potential of RF detection, we believe that the accuracy performance of RF detection is still under-investigated. Most of the works on RF detection evaluate drone detection and classification using limited datasets. The experiments used in these studies are also performed in restricted and semi-controlled environment. On the contrary, the performance of RF-detection in real-world deployment heavily depends upon the interference from other RF sources and the impact of rugged terrain on the drones' RF signals. We believe that RF detection is still an area which requires more intensive research and experimental field validations. RF detection can be applied to multiple drone detection, although none of the existing studies investigate this.

C. Radar-Based Detection

Radars (Radio Detection and Ranging) are traditionally being used for detecting aircraft in military and civil applications (e.g., aviation) and thus are generally considered as a proven tool for detecting drones as well. A radar transmits a radio wave (typically in microwave frequency range) and receives the reflected wave from an object (i.e., aircraft or

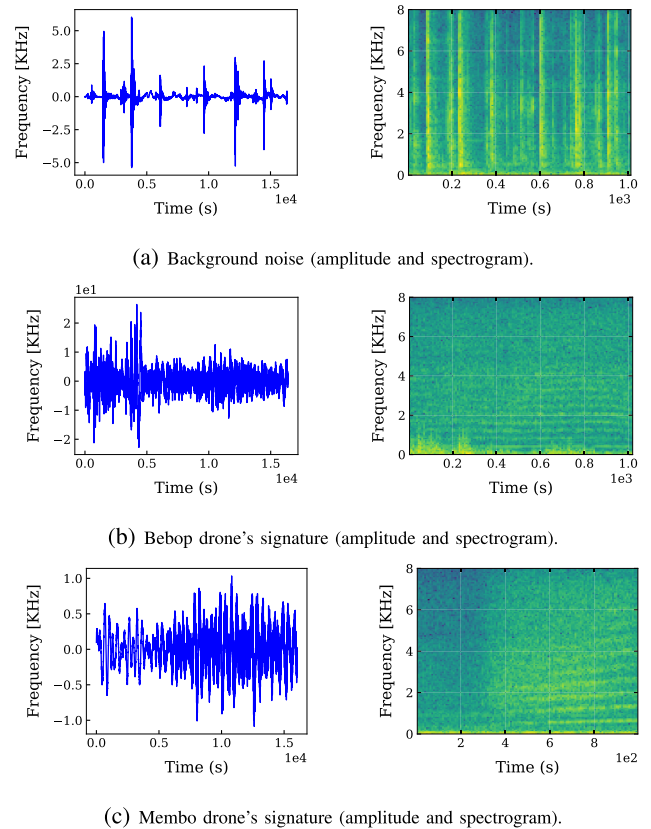


Fig. 5. Sample acoustic signatures of background noise and two commercially available drone models (i.e., Bebop, Membo by Parrot) extracted from dataset [27].

drone). The receiver analyzes the Doppler shift caused by the moving object to detect the presence, distance, and speed of the object. Radars can be categorized as passive or active radars. Active radar [71] transmits a signal and received the reflected signal to detect objects. On the contrary, a passive

TABLE III
COMPARISON OF DRONE DETECTION METHODS

Method	Input	Typical Range	LOS Required	Accuracy	Capabilities	Performance deteriorating factors
Acoustic detection	Sound wave produced by drone's motor ($\sim 40\text{KHz}$)	$\sim 200\text{ m}$	No	Low	Detection, identification, localization, payload detection	Background noise
RF detection	Radio communication between drone and controller	$\sim 1\text{ km}$	No	High	Detection, identification, localization	Harsh Weather, RF interference, Autonomous drones
Radar detection	High-frequency waves reflected from drone.	$\sim 100\text{ km}$	No	High	Detection, identification, localization, tracking	Lower-sized drones, mistaken by stationary objects & birds
Visual detection	Video/imagery of drones using camera	$\sim 100\text{ m}$	Yes	High	Detection, identification, tracking, payload detection	Low light conditions, dust, fog etc.

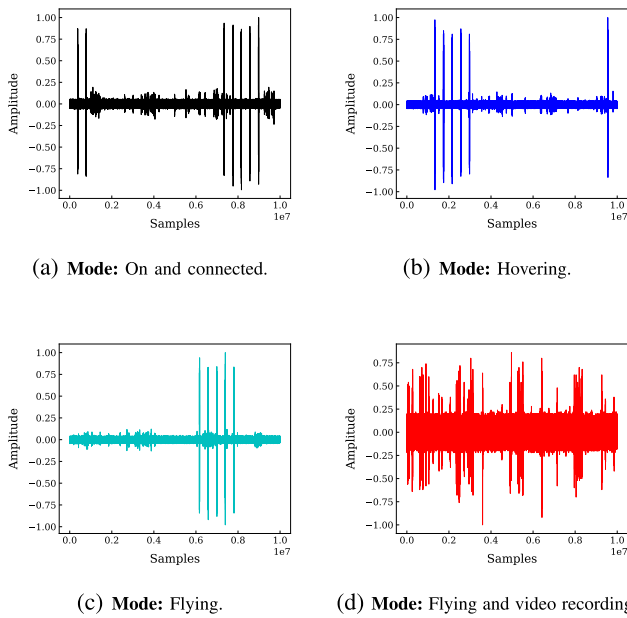


Fig. 6. RF signature of Bebop drone in various activities/modes (samples taken from DroneRF dataset [55]).

radar [72] does not transmit any signal itself and rely on other signal sources such as analogue television, FM radio, cellular signals, and digital audio and video broadcasting etc., Active radars are typically used for drone detection and generally fall into two categories: *pulse radar* and *Frequency Modulated Continuous Wave (FMCW) radar*. Pulse radars transmit short pulses and measure the time delay between the transmitted and received signals to detect objects whereas, FMCW radars transmit continuous waves and measure the difference in the frequency of transmitted and received signals to determine distance to the object. Some commercial off-the-shelf radar based drone detection systems include Kaspersky radar system [73], Echodyne's radar system [74], and DeTect radar systems [75].

1) Advantages and Limitations: The major advantage of radar-based detection is the high detection and localization accuracy as compared to any other method. Unlike acoustic or RF detection, radars are capable to detect autonomous drones and drones with noise suppression. The detection accuracy

is usually not impacted by visual conditions such as rain, fog, and dust etc. Thus, radars are typically considered as optimal choice when high accuracy and reliability is desired. However, radars do have few intrinsic limitations. First, traditional radars are designed to detect aircrafts with large radar cross-section (RCS) and fall short in detecting smaller objects like drones. Second, radars may not differentiate between a hovering (not changing location) or slowly moving drones and a static reflective object. Third, radars also often fail to distinguish between small-sized commercial drones and birds. They also poorly perform in detecting many drones flying in a swarm erratically at varying speeds. Fourth, radars incur high cost of deployment. Radars also use high frequencies which require regulatory licence; making radars unsuitable for drone detection in many applications. The inherited advantages and limitations are also associated to the type of radars i.e., pulse and FMCW. Pulse radars are usually more expensive to build due to the complexity involved in measuring the very short amount of time. They are also less accurate at shorter ranges. On the other hand, FMCW radars are less expensive. Pulse radars are good to measure the speed of the moving target, which is challenging for FMCW radars. On the other hand, FMCW radars have relatively better range resolution (up to 0.5m) and are more accurate at shorter distances. Pulse radars usually become less accurate at shorter distances.

2) State-of-the-Art: Radars were originally designed to detect large-sized aircrafts and tends to fail sometime to accurately detect small size drones. A micro-Doppler radar can detect small Doppler shifts generated by internal parts of drones. A micro-Doppler radar is proposed in [76] using a boosting classifier to distinguish between drones (four) and a birds (three). The proposed radar uses six features extracted from Time Velocity Diagrams (TVDs). The features also known as the Kim-Ling features [77] are base velocity, bandwidth of Doppler signal, offset, bandwidth without micro-Doppler, normalized standard deviation of Doppler signal strength, and cycle frequency. In [78], a drone localization and activity classification method is proposed using mmWave Frequency Modulated Continuous Wave (FMCW) radar. The drone's location is estimated from the range and elevation angle of arrival, whereas the micro-Doppler signatures are used to classify drone's activity by applying ML techniques

(SVM, CNN and GBM), reportedly best results achieved using CNN. Authors in [79] conducted field tests to demonstrate a Persistent FMCW radar system to detect a single UAV. The system use Cell Averaging Constant False Alarm Rate (CA-CFAR) and achieved a range of up to 2 Km in outdoor environment containing clutter. In [80], authors presented a method for drone detection and classification leveraging the RCS signatures. However, the work is limited to simulation analysis and experimental details are limited.

Although RCS signatures can be used to implement drone detection and classification, it requires upgrading the radar calibration and other configurations. Authors in [81] proposed an independent method which does not uses RCS signatures. The method uses flight mechanic and behavior modes of drones and birds represented as discrete motion characteristics and derive five features i.e., mean speed, Standard Deviation (SD) of speed, SD of heading, maneuverability factor, and oscillation factor. Some similar works on the use of radars for drone detection can be found in [82] and [76], [83]–[89]. Authors in [90] propose to deploy radars on one drone to track other unauthorized drone. However, this is based on the assumption that there is only one intruder drone which is not a practical assumption.

3) Summary and Insights: Radar-based moving object detection and tracking is a mature technology providing accurate detection and location information of moving objects such as aircrafts. However, it faces challenges when applied to detect smaller objects such as drones. The review of the state-of-the-art shows that FMCW radars are mostly applied for drone detection as FMCW radars are more accurate at shorter distances (50-100m) at which pulse radars typically becomes blind. Although Pulse radars can be relatively more accurate than FMCW radar at long range, the hardware complexity of pulse radars would hinder its adoption. Both pulse and FMCW radars can detect multiple drones.

While the technical challenges can be overcome with state-of-the-art approaches such as micro-Doppler, the regulatory requirements will hinder its wide-scale deployment particularly in civil applications. Apart from that, radars are also difficult to deploy and require high transmit power (for detection over longer ranges). They also require more space, and the maintenance cost is high. Hence, radars would not be well-perceived for drone detection in many civilian applications.

D. Visual Detection

Camera-based visual detection schemes employ computer vision and deep learning to detect drones using cameras. A visual detection system typically uses daylight cameras and infrared or thermal imaging [91] techniques to capture drones images or videos and then uses computer vision based object detection techniques [92], [93] to detect drones in them. Fig. 7 shows sample images of a drone model captured using a daylight and thermal imaging.

Two common approaches are used for visual detection of drones in imagery data; (i) conventional techniques, and (ii) DL-based techniques. Conventional detection techniques typically work in two steps. In the first step, images are processed



(a) Sample Image.

(b) Sample Thermal Image.

Fig. 7. Sample images from MCL drone dataset [94].

to manually extract features such as Histogram of Oriented Gradients (HOG) [95], Scale Invariant Feature Transform (SIFT). In the second step, the extracted features are classified using ML techniques such as SVM or boosting algorithms. Unlike the conventional approach, DL-based detection does not use manually extracted hand-crafted features. Instead they rely on more sophisticated techniques such as CNN which automatically extract features in images while training a detection model. Thus, an end-to-end drone detection DL model can be trained using DL with high accuracy. Some commercial off-the-shelf product using visual drone detection include Aerial Lens [96], AARTOS Camera [97], and DEDrone [98].

1) Advantages and Limitations: The use of state-of-the-art DL techniques makes visual detection a more attractive approach. Visual drone detection offers several features such as presence detection, identification, tracking and even payload detection. However, it also comes with challenges. Visual detection requires a clear LOS to the target drone. The detection range is also limited. Longer detection range and 360° of coverage requires the use of multiple high resolution cameras, which will increase the system cost. The accuracy performance degrades in harsh environments such as dust, fog, and low light conditions and may fail completely in the night. Thus an equal number of infrared cameras will be needed to cope with the poor lighting condition, which further increase the system cost. Visual detection requires a clear high resolution image of the target scene. When a drone is moving with very high speed, the drone may look blur and hence difficult to be detected. Also, if the drone is at higher distance, while the camera is of low resolution, the image quality will be low and the drone will appear smaller which will consequently affect the detection accuracy.

2) State-of-the-Art: There are few works that adopted conventional computer vision techniques for drone detection. For instance, [99] propose an identification, localization and tracking system for multi-drones using computer vision. The system uses motion-based blob detection for feature extraction and Kernelized Correlation Filter (KCF) [100] and Discriminative Correlation Filter (DCF) [101] for detection. Field tests were performed over two drones for detection and tracking. Similarly, [102] uses a UAV detection and distance estimation system using computer vision. The authors use hand-crafted features i.e., Haar-like feature, HOG, Local Binary Patterns (LBP), a cascaded boosted classifier to detect drones in images and SVM to estimate distance between camera and the drone. Although not specifically proposed for

TABLE IV
STATE-OF-THE-ART IN DRONE DETECTION

Detection	Ref	Task/Category	Features	Method	Data/Evaluation
Acoustic Detection	[44]	Identification	Spectrograms	Random Forest (RF)	Drone's recordings
	[36]	Detection	PSD	Cross-Correlation	Studio Recording of drone sounds
	[37]	Detection	MFCC, LPCC	SVM	Sounds data from Internet
	[40]	Detection	PSD, RMS, MFCC	SVM, KNN with ICA	Sounds data from Internet
	[41]	Detection	PSD, RMS, MFCC	SVM, KNN with IVA	Sounds data from Internet
	[33]	Detection & localization	MFCC	SVM and TDOA	Outdoor flying drones
	[27]	Detection & classification	Automatic	CNN, RNN	Simulation
	[45]	Identification	MFCC	LSTM	Outdoor flying drones, YouTube
	[42]	Detection	MFCC, Spectrogram	GMM, CNN, RNN	Recording of flying drones
RF-based Detection	[122]	Detection	RF waveforms, PSD	Fourier analysis	Experimental Testbed
	[58]	Detection	Packet level features	RT, RF, SMO, Logit Boost	Experimental Testbed
	[60]	Detection & classification	NCA	KNN, DA, SVM, NN	Monte-carlo simulation
	[66]	Identification	Processed sounds	Deep Learning	Simulation
	[123]	Detection & classification	Spectrograms	CNN	Publicly available dataset.
	[68]	Detection & classification	Spectrograms	CNN	ML, Experimental testbed
	[61]	Detection	SFS, WEE, PSE	SVM, RF, NB, EL, KNN	ML model
	[63]	Detection & Identification	FD, AIB, SIB	PCA, NCA	ML model
	[69]	Identification	Processed sounds	DRNN	DroneRF dataset [66]
Radar Detection	[79]	Detection	NA	CA-CFAR.	Evaluation
	[76]	Detection & classification	Kim-Ling features [77]	Booster, SVM	ML model with field data
	[124]	Detection & tracking	Multi sensor fusion	Time Division Multi Frequency (TDMF)	Simulation & experimentation
	[80]	Detection & classification	RCS signatures	Monte Carlo	Simulation
	[78]	Detection & classification	AoA, microDoppler	SVM, CNN, GBM	Numerical modeling & ML
	[81]	Detection & classification	speed, heading, maneuverability, oscillation	RF	Data from field experiments
Visual Detection	[99]	Detection & tracking	Regions in image	KCF, DCF	Field tests
	[102]	Detection & distance estimation	Haar-like feature, HOG, LBP	Boosting classifier, SVM	Testbed
	[104]	Detection & classification	images	YOLOv3 CNN [115]	Experimental testbed
	[105]	Detection & identification	images	Haar Feature and CNN	Simulation
	[7]	Detection & tracking	video frames	CNN models	Testbed
	[121]	Detection & tracking	images	CNN	Testbed
	[106]	Detection & tracking	video frames	6 CNN models	Testbed

drones, the work in [103] propose fast pixel-wise posterior 3D (fast-PWP3D) algorithm for target tracking using visual sensors.

A review of most recent literature (as in [104]–[114]) shows a growing interest in the use of DL-based visual detection approaches. For instance, authors in [104] use the pre-trained YOLOv3 [115] CNN model. The model is evaluated on a dataset of drone images collected at different timing, scales, and varying altitudes using NVIDIA Jetson TX2. Similarly, [105] trained a simple CNN model over collected drones' imagery for both detection and identification purposes. A real-time drone tracking system using DL is presented in [116]. The system uses five popular CNN models (YOLOv2, MobileNet [117], InceptionV2 [118], R-FCN [119], and R-CNN [120]) trained over a dataset containing images of 11 different drone models. A PTZ camera takes videos of the environment and send it to a workstation. The workstation contains the trained models to detect drones in video frames. When a drone is detected in a frame, the workstation sends a control message to the PTZ camera to track the drone. Authors in [106] provide a more advanced system

for drone detection, localization and tracking using a headset equipped with camera and Inertial Measurement Unit (IMU). The camera captures image streams of a scene which is fed to a CNN model for detecting drone. When a drone is detected, the IMU is used to track the drone. The system does not uses any GNSS for localization.

Most of the works using DL-based visual drone detection schemes use datasets of drone images captured from the ground. Using such datasets are suitable for ground detection system using PTZ cameras. However, using PTZ cameras is not suitable in congested environment where a LOS to the target drone is often lost. An alternate approach is the air-to-air visual detection [121], which uses a drone detection and tracking using another flying drone equipped with a camera and onboard detection system. The existing datasets that typically contain drones images captured from a ground camera may not provide sufficient accuracy due to the fact that drones' appear different when seen from a flying drone. Hence, authors in [121] developed a large dataset i.e., "Det-Fly" containing 13,000 images of drones captured from another flying drones.

TABLE V
A LIST OF PUBLICLY AVAILABLE DATASETS FOR DRONE'S DETECTION

Method	Ref	Dataset	Year	Details
Acoustic Detection	[27]	Sara et al.	2019	1300 sounds samples of 2 drones recorded while flying and hovering in a quite indoor environment and other sounds of non-drone objects (from Internet).
	[45]	DronePrint	2021	Sounds samples of 20 drones and other non-drone objects collected from experiments and online resources.
RF Detection	[55]	DroneRF	2019	227 recorded samples of 3 drones along with recordings of background RF activities.
	[125]	AirID	2020	I/Q samples from UAV RF transmissions.
	[126]	Hovering UAVs	2020	RF samples of 7 hovering UAVs.
Visual Detection	[127]	USC drone	2017	30 visible video clips of a single drone model. Each video is approximately one minute long, captured at 30fps. Each frame resolution is 1920×1080 .
	[127]	USC thermal drone	2017	10 thermal video clips of the same drone model as in USC drone dataset, with same video format.
	[128]	UAVdata	2021	13,803 images consisting of images of balloons and 6 types of UAVs.
	[94]	MCL Drone	2019	A combination of (i) USC drone dataset, (ii) USC thermal drone dataset and (iii) Public-Domain drone dataset.
Hybrid Detection	[129], [130]	Multi-Sensor dataset	2021	Visible and infrared videos i.e., 365 IR and 285 visible videos (each of 10s) of 3 drones, birds, airplanes and helicopters) and 90 audio files of 3 drones.

3) Summary and Insights: Visual detection despite its inherited limitations (e.g., LOS and proper lighting requirements) is gaining popularity in the academic research likely due to the success and growing interest of researchers in the area of DL and computer vision. Although DL-based visual detection provides a significant improvement to detect drones of various sizes and shapes, there are still various challenges seeking attention. Although several visual detection of drones exists, none of these address the drone localization (determining the exact location i.e., coordinates of the drone). Detecting drone only is not a sufficient feature of a real-world drone detection system as detection is typically followed by disruption (a response/action to capture or destroy the drone) [131]. Without the position information of the detected drone, disruption is not possible.

Another key issue is the unreliable accuracy of visual detection that significantly deteriorate when environmental conditions worsen e.g., in poor light, dust, fog etc., [132]. If cost is not a primary constraint, the incorporation of infrared cameras can be used to cope with poor light conditions. Moreover, conditions such as heavy fog or dust are still challenging that will cause worse performance at longer distances. Nonetheless, visual detection seems potentially more attractive approach due to the ease of deployment, affordability of high resolution cameras, and the promising applications of edge computing [133], [134] to run DL models at the edge devices. Visual detection is also the most suitable method for detection multiple drones and can detect accurately a large number of drones with accurate count using high resolution images without additional cost of the system.

A brief comparison of the four detection approaches is presented in Table III. A summary of the state-of-the-art is presented in Table IV. We also studied various publicly available datasets and provided their summarised details in Table V that can be further explored to evaluate the various detection schemes.

IV. SENSOR FUSION

The analysis of the four detection approaches discussed in the earlier section reveals that none of the methods alone solves the drone detection problem in all scenarios. Hence, other methods based on sensor fusion are being developed to overcome the limitations of the individual methods and to accomplish the detection tasks with high accuracy and reliability. There are two main approaches for sensor fusion based detection as depicted in Fig. 8. In the first method, two or more different sensors work in parallel to detect drones. The output of both sensors is then combined in two possible ways i.e., (i) data collected by both sensors (or the features extracted from the data) is combined and fed to a single detection system), or (ii) the detection output of both sensors (decision) are combined (averaged or weighted averaged) to trigger the final detection. The works that fall in this category include [135]. In [135], authors propose a system that consists of 3 microphone (acoustic sensors) and 30 cameras (visual sensors) which constantly monitor a given area and generates audio and video streams. Both audio and video streams are then processed to extract features which are then fed to a classifier to perform the detection. The results show the audio-assisted system (95.74%) significantly improves the performance of the visual detection

TABLE VI
METHODS USING SENSOR FUSION AND ALTERNATE TECHNIQUES FOR DRONE DETECTION

Method	Ref	Task	Description of work	Limitations
Visual & acoustic sensors fusion	[135]	Detection	Use of acoustic sensors to overcome occlusions in visual detection.	Increased system complexity and cost.
	[130]	Detection & tracking	A system of acoustic sensors and visible & IR cameras to improve accuracy.	Sensor fusion approach is simple and not unified.
Radar & visual sensor fusion	[136]	Detection	Use of acoustic sensors to overcome occlusions in visual detection.	Increased system complexity and deployment cost.
	[137]	Detection	Similar work in [136] on sensor fusion.	High complexity and cost.
RF & visual sensor fusion	[138]	Detection	DL-based detection using combined features from RF signals and images.	Synthetic image images used due to unavailability of images for the original RF signatures.
LED	[9]	Detection	Spying drone detection using LED watermark detection in drone aided captured videos.	System complexity of water mark projection is impractical.
RFID	[141]	Identification	RFID-based drone identification	Limited range, low accuracy, require LOS.
	[140]	Localization	RFID & RF fusion for drone detection	Can not be used to detect drones but to find location only.
Laser	[139]	Detection	Laser (LiDAR)-based drone detection	Does not provide identification and need LOS. Mistake drone with other objects (e.g., birds)

system (79.54%). Another work in [130] develop a drone detection and tracking system with sensor fusion extension to perform sophisticated detection decision based on outputs of multiple sensors (acoustic sensors, thermal cameras, and daylight cameras) with the possibility to configure sensors selection and weighted outputs of the selected sensors. However, with widely available settings of the sensors selection and fusion, the accuracy performance are not evaluated for hybrid detection. Similar to [130], [135], authors in [136] and [137] propose fusion of visual and radar detection. A Scanning Surveillance Radar System (SSRS) aided with camera is used to detect a drone using radar whereas its identification is further confirmed by the camera. The authors present the system design without detailed investigation of the performance of such system. Fusion of RF and visual sensors is proposed in [138]. The RF signal and the images are processed using Feed-forward Neural Network (FNN) and CNN respectively, to extract features which are then fed to another Deep Neural Network (DNN) to trigger the detection alarm. The work reported an accuracy of 75% using the RF-visual sensor fusion. However, the accuracy of the system is not compared against detection using individual sensors.

In the second method, one sensing method provide a relatively less accurate early detection which triggers the second sensor to adjust its parameters and verify the detection with high reliability. For instance, one can use an RF sensor to detect the presence of a drone at a longer distance with an estimation of the angle of arrival (AoA). The information is then fed to the visual detection part and the camera rotates in the direction of the arriving drone and adjust its zoom level based on the distance from the camera. The drawback of this

approach is that if the information provided by the RF sensor is not accurate, the camera won't be able to perform accurate detection.

In addition to sensor fusion, some researchers adopted alternate methods for detecting drones using other techniques such as laser, Light Emitting Diode (LED) light, and Radio Frequency Identification (RFID) etc. In [9], authors propose a system to detect spying drones using watermark detection in the video captured by the drone. The system uses applies a periodic stimulus using LED lights which appears in any video captured by a spying drone. The video traffic sent by drone to the controller is intercepted and analyzed for the presence of the watermark. Drone detection using 3D laser sensor i.e., LiDAR has been proposed in [139]. The proposed system can detect drones in the range of 2 km even in low light at night. However, LiDAR-based system can not differentiate among drones and birds which is a major drawback. RFID-based detection methods have been proposed earlier in [140] and [141]. However, these methods are not very popular due to limited detection and localization accuracy and requirement of LOS. A summary of methods using sensor fusion and alternate methods is presented in [Table VI](#).

A. Summary and Insights

Sensor fusion is a promising approach to build more reliable, robust, and customized systems to detect drones in various scenarios. A general trend in sensor fusion is to use visual detection in combination with acoustic/RF/radar detection. A more common approach for sensor fusion is using two sensing technologies and their combined outputs to be used for detection decision. An alternative approach is to

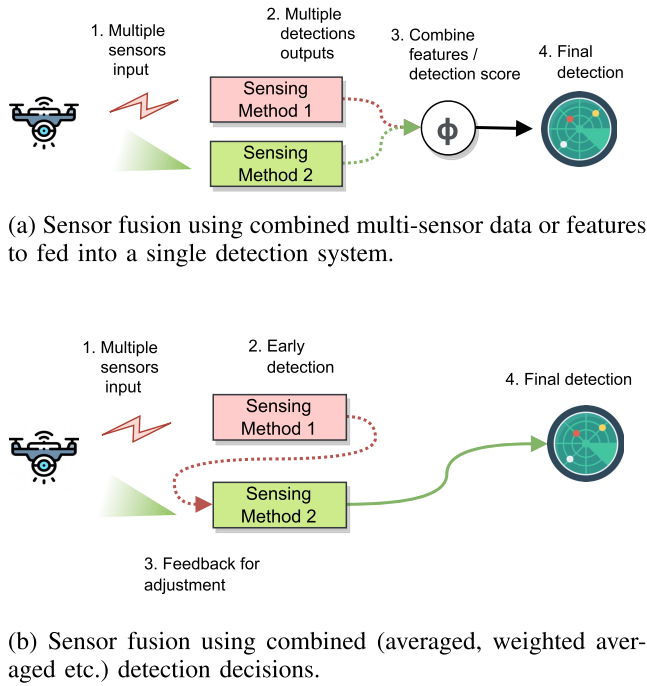


Fig. 8. Hybrid detection using sensors fusion.

use a long range, NLOS detection scheme (e.g., acoustic, RF, or radar) to generate an early detection alarm and then use its output to trigger the second sensor (typically camera) to adjust its configurations such as direction, zoom level etc. to perform a more reliable and accurate detection. Although sensor fusion is a promising approach, there need a significant research and experimental validation to develop candidate systems to serve particular use cases. The key drawbacks of sensors fusion include higher system complexity, computational complexity, and deployment cost. When multiple sensors are integrated to perform joint detection, issues related to synchronisation and delay can arise. For instance, when the output of the RF detection stage is fed as input to a camera for final detection (as illustrated in Fig. 8(a), the RF detection must be completed within a tight delay deadline so that the camera rotates to the target's current location. If the target is moving at higher speed that the additional programming may be required to calculate the updated target location from the predicted location (using RF detection). Similarly, when sensor fusion involves to use features extracted from different sensors (e.g., sound signatures, images etc.) in training a joint DL model, the individual sensors must be synchronized in time so to ensure that the the different features represents the same time instance.

V. CONCLUSION AND FUTURE INSIGHTS

To protect people, assets and critical infrastructure from the possible attacks using unauthorized use of civilian drones, an efficient and reliable drone detection method is urgently needed. However, a detailed review of the state-of-the-art reveals that none of the existing drone detection approaches provide sufficient accuracy and reliability for many practical use cases. In fact, the existing approaches have serious limitations to be adopted as a standalone approach for detecting

drones in diverse situations. Consequently, some researchers suggest using scenario-specific approaches whereas others propose hybrid approaches based on sensor fusion to enhance system's performance and reliability. However, there is still a room for investigation for a more accurate and reliable detection system. The general trend is to rely on sensor fusion, but the sensors selection and system configuration is yet an open research area.

We advocate a scenario-specific hybrid detection system that satisfies the minimum scenario requirements. Based on the list of detection factors in Table II and [142], one can choose a minimum set of basic requirements for a specific scenario. In more general cases, we propose a minimum set of three requirements as these are generally employed in almost any detection scenario. These include:

- *Detection range*: This include the horizontal distance and altitude at which the drone shall be accurately detected.
- *Surrounding*: This include the external factors (such as physical obstacles, light, weather conditions, and ambient noise level) at which drone shall be accurately predicted.
- *Identification*: This include the information such as the drone's make and model that shall be predicted with the model (Usually, other drone's capabilities can be typically determined if the drone's make and model is known.).

While none of the four detection systems outperforms others over all the three requirements, we believe that the visual detection approach has a great potential to meet these requirements and is a relatively a more suitable choice in diverse scenarios. It can detect drones at a reasonable distance if the cameras are installed at the boundaries of the region of interest. If longer detection range is desired, cameras can be aided by RF sensors. To cope with the LOS requirements, the camera's locations can be configured to reliably detect drones approaching from any direction. Furthermore, IR cameras can be used to boost the detection accuracy in low light and harsh weather conditions. As far as the identification is concerned, visual detection potentially offers superior performance to provide identification of drones and any payload carried over it. Furthermore, most of the recently developed techniques apply deep learning for higher detection accuracy. However, datasets used by these studies are very limited. We believe that the availability of larger datasets with extensive drone signatures (i.e., sounds, RF, images etc.) shall be collected over time. The availability of large datasets will further verify the performance of the existing techniques and will help to overcome the limitations as well. To this end, we have covered the existing publicly available datasets for each detection approach in this paper. Another interesting area for future research is to build more robust systems by integrating network security such as authentication based on advanced key-encryption techniques with a physical detection method (more likely RF or visual detection). Such methods can be typically less expensive and more reliable than other sensor fusion of two physical detection methods.

REFERENCES

- [1] L. Abualigah, A. Diabat, P. Sumari, and A. H. Gandomi, "Applications, deployments, and integration of Internet of Drones (IoD): A review," *IEEE Sensors J.*, vol. 21, no. 22, pp. 25532–25546, Nov. 2021.

- [2] *Recreational Flyers & Modeler Community-Based Organizations*. Accessed: Dec. 15, 2021. [Online]. Available: https://www.faa.gov/uas/recreational_fliers/
- [3] *FAA Airspace Forecast—Fiscal Year 2019–2039*. Accessed: Dec. 15, 2021. [Online]. Available: https://www.faa.gov/data_research/aviation/aerospace_forecasts/media/FY%2019-39_FAA_Aerospace_Forecast.pdf
- [4] M. B. Cruzan *et al.*, “Small unmanned aerial vehicles (micro-UAVs, drones) in plant ecology,” *Appl. Plant Sci.*, vol. 4, no. 9, Sep. 2016, Art. no. 1600041.
- [5] G. Tian, C. Xiao, and W. Liu, “Dynamic no-fly zone for drones,” in *Proc. IEEE SmartWorld, Ubiquitous Intell. Comput., Adv. Trusted Comput., Scalable Comput. Commun., Cloud Big Data Comput., Internet People Smart City Innov. (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI)*, Aug. 2019, pp. 761–766.
- [6] *The Drone Security Threat*. Accessed: Dec. 15, 2021. [Online]. Available: <https://me-en.kaspersky.com/resource-center/threats/can-drones-be-hacked>
- [7] S.-H. Park and K.-H. Lee, “Developing criteria for invasion of privacy by personal drone,” in *Proc. Int. Conf. Platform Technol. Service (PlatCon)*, Feb. 2017, pp. 1–7.
- [8] Y. Tian *et al.*, “Cost-effective NLOS detection for privacy invasion attacks by consumer drones,” in *Proc. IEEE/AIAA 38th Digit. Avionics Syst. Conf. (DASC)*, Sep. 2019, pp. 1–7.
- [9] R. Ben Netanel, B. Nassi, A. Shamir, and Y. Elovici, “Detecting spying drones,” *IEEE Secur. Privacy*, vol. 19, no. 1, pp. 65–73, Jan. 2021.
- [10] P.-Y. Kong, “A survey of cyberattack countermeasures for unmanned aerial vehicles,” *IEEE Access*, vol. 9, pp. 148244–148263, 2021.
- [11] C. Kumar and S. Mohanty, “Current trends in cyber security for drones,” in *Proc. Int. Carnahan Conf. Secur. Technol. (ICCSST)*, Oct. 2021, pp. 1–5.
- [12] F. Li, X. Wang, H. Chen, K. Sharif, and Y. Wang, “ClickLeak: Keystroke leaks through multimodal sensors in cyber-physical social networks,” *IEEE Access*, vol. 5, pp. 27311–27321, 2017.
- [13] B. Heubl, “Consumer drones used in bomb attacks: Terrorist rebels are turning consumer drones into deadly weapons. E&T investigates why it goes on and what can be done about it,” *Eng. Technol.*, vol. 16, no. 4, pp. 1–9, 2021.
- [14] S. Hameed, F. Junejo, M. A. Y. Zai, and I. Amin, “Prediction of civilians killing in the upcoming drone attack,” in *Proc. IEEE 5th Int. Conf. Eng. Technol. Appl. Sci. (ICETAS)*, Nov. 2018, pp. 1–5.
- [15] *Drone Attacks Target Saudi Airport in Jazan*. Accessed: Jan. 13, 2022. [Online]. Available: <https://www.aljazeera.com/news/2021/10/9/drone-attack-targeting-saudi-airport-leaves-several-injured>
- [16] *Russia Says Mysterious Armed Drones are Attacking its Military Bases in Syria—and They Don’t Know Who’s Sending Them*. Accessed: Dec. 15, 2021. [Online]. Available: <https://www.independent.co.uk/news/world/middle-east/russia-military-bases-drones-syria-armed-attacks-tartus-uavs-latakia-a8151066.html>
- [17] J. S. Patel, F. Fioranelli, and D. Anderson, “Review of radar classification and RCS characterisation techniques for small UAVs or drones,” *IET Radar, Sonar Navigat.*, vol. 12, no. 9, pp. 911–919, Sep. 2018.
- [18] X. Shi, C. Yang, W. Xie, C. Liang, Z. Shi, and J. Chen, “Anti-Drone system with multiple surveillance technologies: Architecture, implementation, and challenges,” *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 68–74, Apr. 2018.
- [19] I. Guvenc, F. Koohifar, S. Singh, M. L. Sichitiu, and D. Matolak, “Detection, tracking, and interdiction for amateur drones,” *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 75–81, Apr. 2018.
- [20] M. M. Azari, H. Sallouha, A. Chiumento, S. Rajendran, E. Vinogradov, and S. Pollin, “Key technologies and system trade-offs for detection and localization of amateur drones,” *IEEE Commun. Mag.*, vol. 56, no. 1, pp. 51–57, Jan. 2018.
- [21] B. Taha and A. Shoufan, “Machine learning-based drone detection and classification: State-of-the-art in research,” *IEEE Access*, vol. 7, pp. 138669–138682, 2019.
- [22] S. Park, H. T. Kim, S. Lee, H. Joo, and H. Kim, “Survey on anti-drone systems: Components, designs, and challenges,” *IEEE Access*, vol. 9, pp. 42635–42659, 2021.
- [23] S. Samaras *et al.*, “Deep learning on multi sensor data for counter UAV applications—A systematic review,” *Sensors*, vol. 19, no. 22, p. 4837, Nov. 2019.
- [24] A. N. Wilson, A. Kumar, A. Jha, and L. R. Cenkeramaddi, “Embedded sensors, communication technologies, computing platforms and machine learning for UAVs: A review,” *IEEE Sensors J.*, vol. 22, no. 3, pp. 1807–1826, Feb. 2022.
- [25] B. Kang, H. Ahn, and H. Choo, “A software platform for noise reduction in sound sensor equipped drones,” *IEEE Sensors J.*, vol. 19, no. 21, pp. 10121–10130, Jul. 2019.
- [26] L. Schauer, F. Dorfmeister, and F. Wirth, “Analyzing passive Wi-Fi fingerprinting for privacy-preserving indoor-positioning,” in *Proc. Int. Conf. Localization GNSS (ICL-GNSS)*, Jun. 2016, pp. 1–6.
- [27] S. Al-Emadi, A. Al-Ali, A. Mohammad, and A. Al-Ali, “Audio based drone detection and identification using deep learning,” in *Proc. 15th Int. Wireless Commun. Mobile Comput. Conf. (IWCMC)*, Jun. 2019, pp. 459–464.
- [28] F. Grondin and F. Michaud, “Lightweight and optimized sound source localization and tracking methods for open and closed microphone array configurations,” *Robot. Auto. Syst.*, vol. 113, pp. 63–80, Mar. 2019.
- [29] *Acoustic Drone Detection System Discovair—Rapid Deployable Cuas Solution*. Accessed: Apr. 13, 2022. [Online]. Available: <https://www.sqhead.com/drone-detection/>
- [30] *Drone Hound—Optinav Drone Detection and Tracking System*. Accessed: Apr. 13, 2022. [Online]. Available: <https://www.optinav.com/drone-hound>
- [31] L. Wang and A. Cavallaro, “Acoustic sensing from a multi-rotor drone,” *IEEE Sensors J.*, vol. 18, no. 11, pp. 4570–4582, Jun. 2018.
- [32] S. V. Sibanyoni, D. T. Ramotsela, B. J. Silva, and G. P. Hancke, “A 2-D acoustic source localization system for drones in search and rescue missions,” *IEEE Sensors J.*, vol. 19, no. 1, pp. 332–341, Jan. 2019.
- [33] Z. Shi, X. Chang, C. Yang, Z. Wu, and J. Wu, “An acoustic-based surveillance system for amateur drones detection and localization,” *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 2731–2739, Mar. 2020.
- [34] F. Christnacher *et al.*, “Optical and acoustical UAV detection,” *Proc. SPIE*, vol. 9988, pp. 83–95, Oct. 2016.
- [35] J. Busset *et al.*, “Detection and tracking of drones using advanced acoustic cameras,” *Proc. SPIE*, vol. 9647, pp. 53–60, Oct. 2015.
- [36] M. Balakin, A. Dvorak, and D. Kurylev, “Real-time drone detection and recognition by acoustic fingerprint,” in *Proc. 5th Sci. School Dyn. Complex Netw. Appl. (DCNA)*, 2021, pp. 44–45.
- [37] M. Z. Anwar, Z. Kaleem, and A. Jamalipour, “Machine learning inspired sound-based amateur drone detection for public safety applications,” *IEEE Trans. Veh. Technol.*, vol. 68, no. 3, pp. 2526–2534, Mar. 2019.
- [38] C. Cortes and V. Vapnik, “Support-vector networks,” *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995.
- [39] N. S. Altman, “An introduction to kernel and nearest-neighbor non-parametric regression,” *Amer. Statistician*, vol. 46, no. 3, pp. 175–185, 1992.
- [40] Z. Uddin, M. Altaf, M. Bilal, L. Nkenyereye, and A. K. Bashir, “Amateur drones detection: A machine learning approach utilizing the acoustic signals in the presence of strong interference,” *Comput. Commun.*, vol. 154, pp. 236–245, Mar. 2020.
- [41] Z. Uddin, J. Nebhen, M. Altaf, and F. A. Orakzai, “Independent vector analysis inspired amateur drone detection through acoustic signals,” *IEEE Access*, vol. 9, pp. 63456–63462, 2021.
- [42] S. Jeon, J.-W. Shin, Y.-J. Lee, W.-H. Kim, Y. Kwon, and H.-Y. Yang, “Empirical study of drone sound detection in real-life environment with deep neural networks,” in *Proc. 25th Eur. Signal Process. Conf. (EUSIPCO)*, Aug. 2017, pp. 1858–1862.
- [43] Y. Seo, B. Jang, and S. Im, “Drone detection using convolutional neural networks with acoustic STFT features,” in *Proc. 15th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Nov. 2018, pp. 1–6.
- [44] N. Siriphun, S. Kashiara, D. Fall, and A. Khurat, “Distinguishing drone types based on acoustic wave by IoT device,” in *Proc. 22nd Int. Comput. Sci. Eng. Conf. (ICSEC)*, Nov. 2018, pp. 1–4.
- [45] H. Kolamunna *et al.*, “DronePrint: Acoustic signatures for open-set drone detection and identification with online data,” *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 5, no. 1, pp. 1–31, Mar. 2021.
- [46] J. Mezei, V. Fiaska, and A. Molnar, “Drone sound detection,” in *Proc. 16th IEEE Int. Symp. Comput. Intell. Informat. (CINTI)*, Nov. 2015, pp. 333–338.
- [47] J. Guo, I. Ahmad, and K. Chang, “Classification, positioning, and tracking of drones by HMM using acoustic circular microphone array beamforming,” *EURASIP J. Wireless Commun. Netw.*, vol. 2020, no. 1, Dec. 2020.

- [48] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [49] Y. LeCun *et al.*, "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, 1989.
- [50] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, pp. 533–536, Oct. 1986.
- [51] C. Amendola, J.-C. Faugere, and B. Sturmfels, "Moment varieties of Gaussian mixtures," *J. Algebr. Statist.*, vol. 7, no. 1, pp. 1–17, Jul. 2016.
- [52] P. Flak, "Drone detection sensor with continuous 2.4 GHz ISM band coverage based on cost-effective SDR platform," *IEEE Access*, vol. 9, pp. 114574–114586, 2021.
- [53] *Drone Detection: Everything you Need to Know: Can Drones be Detected?*. Accessed: Dec. 15, 2021. [Online]. Available: <https://www.911security.com/en-us/knowledge-hub/drone-detection/RF>
- [54] B. Kaplan *et al.*, "Detection, identification, and direction of arrival estimation of drone FHSS signals with uniform linear antenna array," *IEEE Access*, vol. 9, pp. 152057–152069, 2021.
- [55] M. S. Allahham, M. F. Al-Sa'd, A. Al-Ali, A. Mohamed, T. Khattab, and A. Erbad, "DroneRF dataset: A dataset of drones for RF-based detection, classification and identification," *Data Brief*, vol. 26, Oct. 2019, Art. no. 104313.
- [56] *DJI Aerospace*. Accessed: Apr. 13, 2022. [Online]. Available: <https://www.dji.com/aeroscope>
- [57] *AirWarden Drone Detection System*. Accessed: Apr. 13, 2022. [Online]. Available: <https://www.aerodefense.tech/airwarden-drone-detection-system>
- [58] I. Bisio, C. Garibotto, F. Lavagetto, A. Sciarone, and S. Zappatore, "Unauthorized amateur UAV detection based on WiFi statistical fingerprint analysis," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 106–111, Apr. 2018.
- [59] S. Sciancalepore, O. A. Ibrahim, G. Oligieri, and R. Di Pietro, "Detecting drones status via encrypted traffic analysis," in *Proc. ACM Workshop Wireless Secur. Mach. Learn. (WiseML)*, 2019, pp. 67–72.
- [60] M. Ezuma, F. Erden, C. K. Anjinappa, O. Ozdemir, and I. Guvenc, "Micro-UAV detection and classification from RF fingerprints using machine learning techniques," in *Proc. IEEE Aerosp. Conf.*, Mar. 2019, pp. 1–13.
- [61] W. Nie *et al.*, "UAV detection and localization based on multi-dimensional signal features," *IEEE Sensors J.*, vol. 22, no. 6, pp. 5150–5162, Mar. 2022.
- [62] T. K. Ho, "Random decision forests," in *Proc. 3rd Int. Conf. Document Anal. Recognit.*, vol. 1, Aug. 1995, pp. 278–282.
- [63] W. Nie, Z.-C. Han, M. Zhou, L.-B. Xie, and Q. Jiang, "UAV detection and identification based on WiFi signal and RF fingerprint," *IEEE Sensors J.*, vol. 21, no. 12, pp. 13540–13550, Jun. 2021.
- [64] F. R. S. K. Pearson, "LIII. On lines and planes of closest fit to systems of points in space," *London, Edinburgh, Dublin Philosoph. Mag. J. Sci.*, vol. 2, no. 11, pp. 559–572, 1901.
- [65] J. Goldberger, S. T. Roweis, G. E. Hinton, and R. Salakhutdinov, "Neighbourhood components analysis," in *Proc. NIPS*, 2004, pp. 1–8.
- [66] M. F. Al-Sa'd, A. Al-Ali, A. Mohamed, T. Khattab, and A. Erbad, "RF-based drone detection and identification using deep learning approaches: An initiative towards a large open source drone database," *Future Gener. Comput. Syst.*, vol. 100, pp. 86–97, Nov. 2019.
- [67] M. Almasri, "Deep learning for RF-based drone detection and identification using Welch's method," in *Proc. 10th Int. Conf. Data Sci., Technol. Appl.*, 2021, pp. 208–214.
- [68] S. Basak, S. Rajendran, S. Pollin, and B. Scheers, "Drone classification from RF fingerprints using deep residual nets," in *Proc. Int. Conf. Commun. Syst. Netw. (COMSNETS)*, Jan. 2021, pp. 548–555.
- [69] A. Gumaei *et al.*, "Deep learning and blockchain with edge computing for 5G-enabled drone identification and flight mode detection," *IEEE Netw.*, vol. 35, no. 1, pp. 94–100, Jan. 2021.
- [70] S. Yang, Y. Luo, W. Miao, C. Ge, W. Sun, and C. Luo, "RF signal-based UAV detection and mode classification: A joint feature engineering generator and multi-channel deep neural network approach," *Entropy*, vol. 23, no. 12, p. 1678, Dec. 2021.
- [71] R. Doviak, D. Zrnić, and D. Sirmans, "Doppler weather radar," *Proc. IEEE*, vol. 67, no. 11, pp. 1522–1553, Nov. 1979.
- [72] H. Kuschel, D. Cristallini, and K. E. Olsen, "Tutorial: Passive radar tutorial," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 34, no. 2, pp. 2–19, Feb. 2019.
- [73] *Kaspersky Radar-Based Drone Detection System*. Accessed: Apr. 13, 2022. [Online]. Available: <https://antidrone.kaspersky.com/en/solution/hardware/radar/>
- [74] *ESA Radar for Counter-UAS & Ground-Based or Airborne DAA (Detect- and-Avoid)*. Accessed: Apr. 13, 2022. [Online]. Available: <https://www.echodyne.com/>
- [75] *Drone Detection Systems & Tracking Radars; BVLOS & Sense-and-Avoid Technology*. Accessed: Apr. 13, 2022. [Online]. Available: <https://detect-inc.com/>
- [76] S. Bjorklund, "Target detection and classification of small drones by boosting on radar micro-Doppler," in *Proc. 15th Eur. Radar Conf. (EuRAD)*, Sep. 2018, pp. 182–185.
- [77] Y. Kim and H. Ling, "Human activity classification based on micro-Doppler signatures using a support vector machine," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 5, pp. 1328–1337, May 2009.
- [78] P. K. Rai *et al.*, "Localization and activity classification of unmanned aerial vehicle using mmWave FMCW radars," *IEEE Sensors J.*, vol. 21, no. 14, pp. 16043–16053, Jul. 2021.
- [79] A. D. de Quevedo, F. I. Urzaiz, J. G. Menoyo, and A. A. Lopez, "Remotely piloted aircraft detection with persistent radar," in *Proc. 15th Eur. Radar Conf. (EuRAD)*, Sep. 2018, pp. 317–320.
- [80] V. Semkin, M. Yin, Y. Hu, M. Mezzavilla, and S. Rangan, "Drone detection and classification based on radar cross section signatures," in *Proc. Int. Symp. Antennas Propag. (ISAP)*, Jan. 2021, pp. 223–224.
- [81] J. Liu, Q. Y. Xu, and W. S. Chen, "Classification of bird and drone targets based on motion characteristics and random forest model using surveillance radar data," *IEEE Access*, vol. 9, pp. 160135–160144, 2021.
- [82] M. Jian, Z. Lu, and V. C. Chen, "Drone detection and tracking based on phase-interferometric Doppler radar," in *Proc. IEEE Radar Conf. (RadarConf)*, Apr. 2018, pp. 1146–1149.
- [83] M. Schneebeli *et al.*, "Drone detection with a multistatic C-band radar," in *Proc. 21st Int. Radar Symp. (IRS)*, Jun. 2021, pp. 1–10.
- [84] S. A. Musa, R. A. Rsa, A. Sali, A. Isma'il, and N. E. A. Rashid, "DVBS based forward scattering radar for drone detection," in *Proc. 20th Int. Radar Symp. (IRS)*, Jun. 2019, pp. 1–8.
- [85] Y. D. Zhang, X. Xiang, Y. Li, and G. Chen, "Enhanced micro-Doppler feature analysis for drone detection," in *Proc. IEEE Radar Conf. (RadarConf)*, May 2021, pp. 1–4.
- [86] G. Fang, J. Yi, X. Wan, Y. Liu, and H. Ke, "Experimental research of multistatic passive radar with a single antenna for drone detection," *IEEE Access*, vol. 6, pp. 33542–33551, 2018.
- [87] M. Messina and G. Pinelli, "Classification of drones with a surveillance radar signal," in *Computer Vision Systems*, D. Tzovaras, D. Giakoumis, M. Vincze, and A. Argyros, Eds. Cham, Switzerland: Springer, 2019, pp. 723–733.
- [88] E. Kaya and G. B. Kaplan, "Neural network based drone recognition techniques with non-coherent S-band radar," in *Proc. IEEE Radar Conf. (RadarConf)*, May 2021, pp. 1–6.
- [89] T. Quilter and C. Baker, "The application of staring radar to the detection and identification of small unmanned aircraft systems in Monaco," in *Proc. 18th Int. Radar Symp. (IRS)*, Jun. 2017, pp. 1–9.
- [90] S. Dogru and L. Marques, "Pursuing drones with drones using millimeter wave radar," *IEEE Robot. Autom. Lett.*, vol. 5, no. 3, pp. 4156–4163, Jul. 2020.
- [91] W. K. Wong, P. N. Tan, C. K. Loo, and W. S. Lim, "An effective surveillance system using thermal camera," in *Proc. Int. Conf. Signal Acquisition Process.*, Apr. 2009, pp. 13–17.
- [92] Z.-Q. Zhao, P. Zheng, S.-T. Xu, and X. Wu, "Object detection with deep learning: A review," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 11, pp. 3212–3232, Nov. 2019.
- [93] P. Tang, C. Wang, X. Wang, W. Liu, W. Zeng, and J. Wang, "Object detection in videos by high quality object linking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 5, pp. 1272–1278, Apr. 2019.
- [94] Y. Wang, Y. Chen, J. Choi, and C.-C.-J. Kuo, "Towards visible and thermal drone monitoring with convolutional neural networks," *APSIPA Trans. Signal Inf. Process.*, vol. 8, no. 1, pp. 1–13, 2019.
- [95] W. Zhou, S. Gao, L. Zhang, and X. Lou, "Histogram of oriented gradients feature extraction from raw Bayer pattern images," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 67, no. 5, pp. 946–950, May 2020.
- [96] *Aerial Lens Drone Detection Camera Systems*. Accessed: Apr. 13, 2022. [Online]. Available: <https://www.aerialarmor.com/drone-detection-equipment/drone-detection-camera>
- [97] *AARTOS Drone Detection*. Accessed: Apr. 13, 2022. [Online]. Available: <https://drone-detection-system.com/aartos-dds/>
- [98] *Dedrone Drone Detection System*. Accessed: Apr. 13, 2022. [Online]. Available: <https://www.dedrone.com/products/counter-drone-solution>

- [99] N. J. Sie, S. Srigrarom, and S. Huang, "Field test validations of vision-based multi-camera multi-drone tracking and 3D localizing with concurrent camera pose estimation," in *Proc. 6th Int. Conf. Control Robot. Eng. (ICCRE)*, Apr. 2021, pp. 139–144.
- [100] Y. Yang and G.-A. Bilodeau, "Multiple object tracking with kernelized correlation filters in urban mixed traffic," in *Proc. 14th Conf. Comput. Robot. Vis. (CRV)*, May 2017, pp. 209–216.
- [101] T. Xu, Z.-H. Feng, X.-J. Wu, and J. Kittler, "Learning adaptive discriminative correlation filters via temporal consistency preserving spatial feature selection for robust visual object tracking," *IEEE Trans. Image Process.*, vol. 28, no. 11, pp. 5596–5609, Nov. 2019.
- [102] F. Gökçe, G. Üçoluk, E. Şahin, and S. Kalkan, "Vision-based detection and distance estimation of micro unmanned aerial vehicles," *Sensors*, vol. 15, no. 9, pp. 23805–23846, Sep. 2015.
- [103] Y. Liu, P. Sun, and A. Namiki, "Target tracking of moving and rotating object by high-speed monocular active vision," *IEEE Sensors J.*, vol. 20, no. 12, pp. 6727–6744, Jun. 2020.
- [104] D. T. Wei Xun, Y. L. Lim, and S. Srigrarom, "Drone detection using YOLOv3 with transfer learning on NVIDIA Jetson TX2," in *Proc. 2nd Int. Symp. Instrum., Control, Artif. Intell., Robot. (ICA-SYMP)*, Jan. 2021, pp. 1–6.
- [105] D. Lee, W. Gyu La, and H. Kim, "Drone detection and identification system using artificial intelligence," in *Proc. Int. Conf. Inf. Commun. Technol. Conver. (ICTC)*, Oct. 2018, pp. 1131–1133.
- [106] M. Pavliv, F. Schiano, C. Reardon, D. Floreano, and G. Loianno, "Tracking and relative localization of drone swarms with a vision-based headset," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 1455–1462, Apr. 2021.
- [107] M. Nalamati, A. Kapoor, M. Saqib, N. Sharma, and M. Blumenstein, "Drone detection in long-range surveillance videos," in *Proc. 16th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Sep. 2019, pp. 1–6.
- [108] A. Carrio, S. Vemprala, A. R. Sanchez, S. Saripalli, and P. Campoy, "Drone detection using depth maps," in *Proc. IEEE Int. Conf. Intell. Robots Syst.*, Oct. 2018.
- [109] R. Hamatapa and C. Vongchumyen, "Image processing for drones detection," in *Proc. 5th Int. Conf. Eng., Appl. Sci. Technol. (ICEAST)*, Jul. 2019, pp. 1–4.
- [110] A. Y. Husodo, G. Jati, N. Alfiany, and W. Jatmiko, "Intruder drone localization based on 2D image and area expansion principle for supporting military defence system," in *Proc. IEEE Int. Conf. Commun., Netw. Satell. (Comnetsat)*, Aug. 2019, pp. 35–40.
- [111] D.-H. Lee, "CNN-based single object detection and tracking in videos and its application to drone detection," *Multimedia Tools Appl.*, vol. 80, pp. 1–12, Nov. 2021.
- [112] Y. Liu *et al.*, "Trajectory and image-based detection and identification of UAV," *Vis. Comput.*, vol. 37, no. 7, pp. 1769–1780, Jul. 2021.
- [113] M. Wu, W. Xie, X. Shi, P. Shao, and Z. Shi, "Real-time drone detection using deep learning approach," in *Proc. MLICOM*, 2018, pp. 22–32.
- [114] E. Unlu, E. Zenou, N. Riviere, and P.-E. Dupouy, "Deep learning-based strategies for the detection and tracking of drones using several cameras," *IPSI Trans. Comput. Vis. Appl.*, vol. 11, no. 1, Dec. 2019.
- [115] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
- [116] J. Park, D. H. Kim, Y. S. Shin, and S.-H. Lee, "A comparison of convolutional object detectors for real-time drone tracking using a PTZ camera," in *Proc. 17th Int. Conf. Control, Autom. Syst. (ICCAS)*, Oct. 2017, pp. 696–699.
- [117] A. G. Howard *et al.*, "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.
- [118] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, vol. 37, Jul. 2015, pp. 448–456.
- [119] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object detection via region-based fully convolutional networks," in *Advances in Neural Information Processing Systems*, vol. 29, D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, and R. Garnett, Eds. Red Hook, NY, USA: Curran Associates, 2016.
- [120] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.
- [121] Y. Zheng, Z. Chen, D. Lv, Z. Li, Z. Lan, and S. Zhao, "Air-to-air visual detection of micro-UAVs: An experimental evaluation of deep learning," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 1020–1027, Apr. 2021.
- [122] P. Nguyen, H. Truong, M. Ravindranathan, A. Nguyen, R. Han, and T. Vu, "Matthan: Drone presence detection by identifying physical signatures in the Drone's RF communication," in *Proc. 15th Annu. Int. Conf. Mobile Syst., Appl., Services*, New York, NY, USA, Jun. 2017, pp. 211–224.
- [123] M. Mokhtari, J. Bajcetic, B. Szadic-Jotic, and B. Pavlovic, "RF-based drone detection and classification system using convolutional neural network," in *Proc. 29th Telecommun. Forum (TELFOR)*, Nov. 2021, pp. 1–4.
- [124] M. Lu, J. Yi, X. Wan, and K. Shu, "Target tracking in Time-Division-Multifrequency-Based passive radar," *IEEE Sensors J.*, vol. 20, no. 8, pp. 4382–4394, Apr. 2020.
- [125] S. Mohanti, N. Soltani, K. Sankhe, D. Jaisinghani, M. Di Felice, and K. Chowdhury, "AirID: Injecting a custom RF fingerprint for enhanced UAV identification using deep learning," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2020, pp. 370–378.
- [126] N. Soltani, G. Reus-Muns, B. Salehi, J. Dy, S. Ioannidis, and K. Chowdhury, "RF fingerprinting unmanned aerial vehicles with non-standard transmitter waveforms," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 15518–15531, Dec. 2020.
- [127] Y. Chen, P. Aggarwal, J. Choi, and C.-C.-J. Kuo, "A deep learning approach to drone monitoring," in *Proc. Asia-Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, Dec. 2017, pp. 686–691.
- [128] Y. Zeng, Q. Duan, X. Chen, D. Peng, Y. Mao, and K. Yang, "UAVData: A dataset for unmanned aerial vehicle detection," *Soft Comput.*, vol. 25, no. 7, pp. 5385–5393, Apr. 2021.
- [129] F. Svanström, F. Alonso-Fernandez, and C. Englund, "A dataset for multi-sensor drone detection," *Data Brief*, vol. 39, Dec. 2021, Art. no. 107521.
- [130] F. Svanstrom, C. Englund, and F. Alonso-Fernandez, "Real-time drone detection and tracking with visible, thermal and acoustic sensors," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 7265–7272.
- [131] D. He, H. Liu, S. Chan, and M. Guizani, "How to govern the non-cooperative amateur drones?" *IEEE Netw.*, vol. 33, no. 3, pp. 184–189, May 2019.
- [132] L. Wang *et al.*, "A smart droplet detection approach with vision sensing technique for agricultural aviation application," *IEEE Sensors J.*, vol. 21, no. 16, pp. 17508–17516, Aug. 2021.
- [133] M. Ishtiaq, N. Saeed, and M. A. Khan, "Edge computing in IoT: A 6G perspective," 2021, *arXiv:2111.08943*.
- [134] D. Unal, M. Hammoudeh, M. A. Khan, A. Abuarqoub, G. Epiphaniou, and R. Hamila, "Integration of federated machine learning and blockchain for the provision of secure big data analytics for Internet of Things," *Comput. Secur.*, vol. 109, Oct. 2021, Art. no. 102393.
- [135] H. Liu, Z. Wei, Y. Chen, J. Pan, L. Lin, and Y. Ren, "Drone detection based on an audio-assisted camera array," in *Proc. IEEE 3rd Int. Conf. Multimedia Big Data (BigMM)*, Apr. 2017, pp. 402–406.
- [136] M. Caris, W. Johannes, S. Stanko, and N. Pohl, "Millimeter wave radar for perimeter surveillance and detection of MAVs (micro aerial vehicles)," in *Proc. 16th Int. Radar Symp. (IRS)*, Jun. 2015, pp. 284–287.
- [137] M. Caris, S. Stanko, W. Johannes, S. Sieger, and N. Pohl, "Detection and tracking of micro aerial vehicles with millimeter wave radar," in *Proc. 46th Eur. Microw. Conf. (EuMC)*, Oct. 2016, pp. 1553–1555.
- [138] M. Aledhari, R. Razzak, R. M. Parizi, and G. Srivastava, "Sensor fusion for drone detection," in *Proc. IEEE 93rd Veh. Technol. Conf. (VTC-Spring)*, Apr. 2021, pp. 1–7.
- [139] B. Kim, D. Khan, C. Bohak, W. Choi, H. Lee, and M. Kim, "V-RBNN based small drone detection in augmented datasets for 3D LADAR system," *Sensors*, vol. 18, no. 11, p. 3825, Nov. 2018.
- [140] J. S. Choi, B. R. Son, H. K. Kang, and D. H. Lee, "Indoor localization of unmanned aerial vehicle based on passive UHF RFID systems," in *Proc. 9th Int. Conf. Ubiquitous Robots Ambient Intell. (URAI)*, Nov. 2012, pp. 188–189.
- [141] A. Buffi, P. Nepa, and R. Cioni, "SARFID on drone: Drone-based UHF-RFID tag localization," in *Proc. IEEE Int. Conf. RFID Technol. Appl. (RFID-TA)*, Sep. 2017, pp. 40–44.
- [142] M. Schneider, D. Lichte, D. Witte, S. Gimbel, and E. Brucherseifer, "Scenario analysis of threats posed to critical infrastructures by civilian drones," in *Proc. 31st Eur. Saf. Rel. Conf. (ESREL)*, 2021, pp. 520–527.



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